Supplementary Information

A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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S1 Mesh Sensitivity of the Segregation Profiles and the Finite-Efficiency Sink Condition

In an earlier version of the model the point-defect site fractions at the grain boundary were fixed at their thermal-equilibrium values,

$$C_{\mathrm{V}}|_{x=0} = C_{\mathrm{V}}^{\mathrm{eq}}, \qquad C_{\mathrm{I}}|_{x=0} = C_{\mathrm{I}}^{\mathrm{eq}}, \tag{S1}$$

a Dirichlet condition that treats the boundary as a perfect sink. With this condition the solution does not converge under mesh refinement. As the grid spacing Δx is reduced, the point-defect profiles steepen at the boundary, and both the gradient and the magnitude of the solute segregation increase monotonically with refinement, Fig. S1(a–c). The width of the segregation profiles likewise fails to converge. The same mesh sensitivity has been reported for other Onsager-type RIS models (Oplinger et al. 2026).

To remove this dependence the boundary is treated as a sink of finite efficiency, i.e. a Robin condition on the defect fluxes,

$$\mathbf{J}_{\mathrm{V}}\cdot\mathbf{n}|_{x=0} = k_{\mathrm{V}}^{\mathrm{GB}}\left(C_{\mathrm{V}}|_{x=0} - C_{\mathrm{V}}^{\mathrm{eq}}\right), \tag{S2}$$

$$\mathbf{J}_{\mathrm{I}}\cdot\mathbf{n}|_{x=0} = k_{\mathrm{I}}^{\mathrm{GB}}\left(C_{\mathrm{I}}|_{x=0} - C_{\mathrm{I}}^{\mathrm{eq}}\right), \tag{S3}$$

where $k_{\mathrm{V}}^{\mathrm{GB}}$ and $k_{\mathrm{I}}^{\mathrm{GB}}$ are the grain-boundary sink efficiencies of vacancies and self-interstitials and $\mathbf{n}$ is the unit normal of the boundary plane pointing from the metal towards the boundary, so that the flux is counted positive towards the boundary. Equation (S1) is the limit $k^{\mathrm{GB}} \to \infty$. With the Robin condition the profiles converge under refinement, Fig. S1(d–f). The sink efficiencies and the defect-mediated transport coefficients are calibrated jointly against the magnitude and the width of the measured segregation profiles; the resulting comparison at 0.5 and 3.0 dpa is shown in Fig. 3 of the main text.

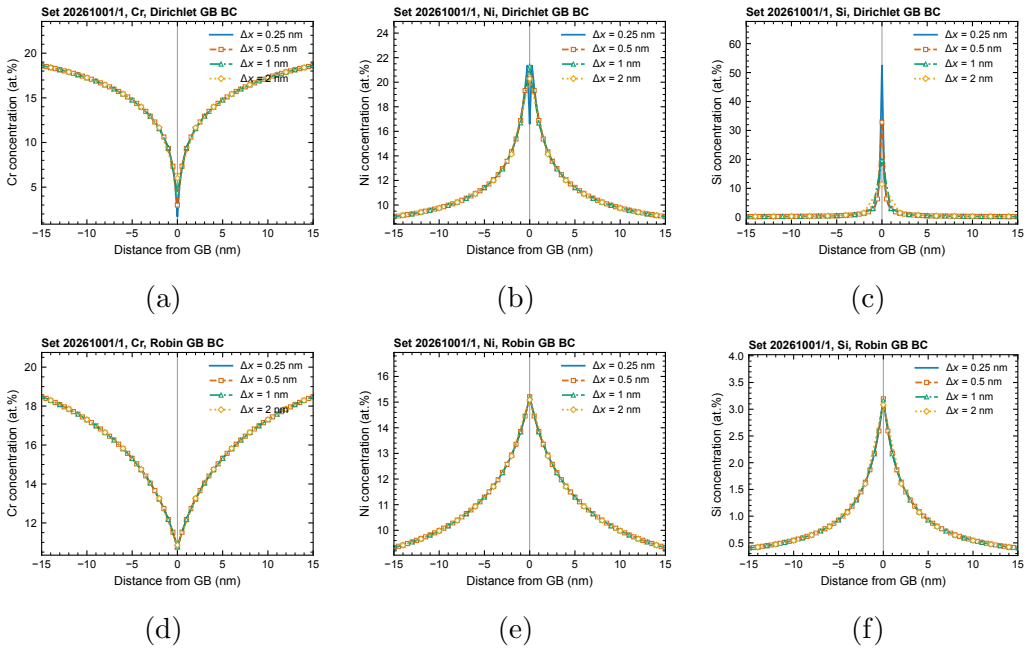


Figure S1: Mesh sensitivity of the calculated Cr, Ni and Si concentration profiles across the grain boundary for the two boundary conditions. (a–c) Dirichlet (perfect-sink) condition, Eq. (S1), at grid spacings $\Delta x = 0.25$, 0.5, 1 and 2 nm. (d–f) Robin (finite-efficiency) condition, Eqs. (S2) and (S3), at the same spacings.

S2 Interfacial Mass Redistribution

S2.1 Motivation

In an earlier version of the model the metal removed at the oxidation front was exactly the metal consumed by the three interfacial reactions, and all unoxidized metal swept by the advancing front was rejected to the metal side of the interface. Because the interfacial rate constant of Cr is much larger than that of Fe, this assumption depletes Cr and enriches Fe strongly in the metal at the interface. The EDX line scans of Deng et al. (2017) show the opposite picture. The oxide is enriched in Cr and depleted in Fe, as expected for selective oxidation, but the metal side of the interface shows no significant Cr depletion or Fe enrichment, only a slight Ni enrichment. In the irradiated specimens the Cr profile has a minimum at the centre of the oxide that deepens with dose, which shows that the interfacial Cr concentration follows the radiation-induced segregation profile as the front advances and is not modulated by the selective oxidation. The excess Fe and Ni must therefore leave the interface by a path other than rejection into the metal. Deng et al. (2020) proposed that they diffuse along the oxide grain boundaries to the surface, where Fe takes part in the formation of the outer oxide particles. A small amount of Ni is detected in the intergranular oxide but no Ni oxide is observed (Deng et al. 2021), so Ni is treated as unoxidized solute, part of which remains in the oxide.

S2.2 Formulation

Let J_{r}^{i} be the rate at which element $i \in \{\mathrm{Cr}, \mathrm{Fe}, \mathrm{Ni}, \mathrm{Si}\}$ is consumed by the interfacial reactions through the stoichiometric sinks of the main text, with $J_{\mathrm{r}}^{\mathrm{Ni}} = 0$, and C_i^{GB} its concentration at the interface. The element with the largest ratio of consumption to availability sets the velocity at which the front must advance into the metal to be supplied,

$$v_{\mathrm{lat}}^{\mathrm{ox,max}} = \max_i \frac{J_{\mathrm{r}}^{i}}{C_i^{\mathrm{GB}}}. \tag{S4}$$

The front sweeps element i at the rate $v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}}$, of which J_{r}^{i} is oxidized. The remainder,

$$E_i = \max\!\left(v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}} - J_{\mathrm{r}}^{i},\, 0\right), \tag{S5}$$

is zero for the element that sets $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ and equals the full swept amount $v_{\mathrm{lat}}^{\mathrm{ox,max}} C_{\mathrm{Ni}}^{\mathrm{GB}}$ for Ni. The excess is partitioned between rejection into the metal and removal through the oxide. The partition is controlled by a redistribution velocity v_i^{r} for each element, the characteristic velocity at which element i can be rejected into the metal ahead of the moving front. The fraction removed through the oxide is

$$\lambda_i = \frac{v_{\mathrm{lat}}^{\mathrm{ox,max}}}{v_{\mathrm{lat}}^{\mathrm{ox,max}} + v_i^{\mathrm{r}}}, \tag{S6}$$

and the net removal of element i from the metal domain is

$$\tilde{J}_{\mathrm{r}}^i = J_{\mathrm{r}}^i + \lambda_i E_i. \tag{S7}$$

$\tilde{J}_{\mathrm{r}}^i$ replaces J_{r}^i as the outgoing flux of element i at the boundary plane and in the recession rate of the interface,

$$v_{\mathrm{ox}} = \sum_i \tilde{J}_{\mathrm{r}}^i = \sum_i J_{\mathrm{r}}^i + \sum_i \lambda_i E_i. \tag{S8}$$

S2.3 Limiting behaviour

- $v_i^{\mathrm{r}} \to \infty$ for every element: $\lambda_i = 0$, the excess is rejected entirely into the metal, $v_{\mathrm{ox}} = \sum_i J_{\mathrm{r}}^i$, and the earlier formulation is recovered.
- $v_i^{\mathrm{r}} \ll v_{\mathrm{lat}}^{\mathrm{ox,max}}$ for every element: $\lambda_i \to 1$, the front outruns the rejection and the whole swept volume is removed. Since $\sum_i C_i^{\mathrm{GB}} = 1$, $v_{\mathrm{ox}} = \sum_i v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}} = v_{\mathrm{lat}}^{\mathrm{ox,max}}$: the interface recedes at the sweep velocity, and the composition profile in the metal is translated rigidly without modification by the selective oxidation.
- Intermediate values give $\sum_i J_{\mathrm{r}}^i \le v_{\mathrm{ox}} \le v_{\mathrm{lat}}^{\mathrm{ox,max}}$, with the partition set element by element.

Equation (S6) is the partition law of nonequilibrium solute capture. In the solute-trapping theory of rapid solidification (Aziz 1982; Aziz and Kaplan 1988) a solute atom swept by the interface escapes into the parent phase only by a diffusive exchange jump of length λ with interface diffusivity D_i; the competition between the advance of the interface by one monolayer, in a time λ/v, and the escape, in a time λ^2/D_i, gives the continuous-growth partition coefficient

$$k(v) = \frac{v/v_D + k_e}{v/v_D + 1}, \qquad v_D = \frac{D_i}{\lambda}, \tag{S9}$$

which goes from the equilibrium value k_e for $v \ll v_D$ to complete trapping, $k = 1$, for $v \gg v_D$. Aziz and Kaplan (1988) note that the model is not specific to solidification and applies to solid-state transformations. Yu et al. (2018) and Sherman et al. (2019) transposed it to the oxide/metal interface as nonequilibrium solute capture: oxides form with compositions outside their equilibrium solubility limits when (i) the captured oxide has a lower free energy than the alloy it forms from and (ii) the effective velocity of cation incorporation exceeds the diffusive velocity of cation exchange across the interface, $v_a/v_D \gg 1$ with $v_D = D_i/a$ and a the hop distance. They argue that for the oxidation of Ni–Cr alloys condition (i) is met over essentially the whole composition range, so that the oxide composition is set by the kinetics, and that condition (ii) holds for most high-temperature and almost all aqueous oxidation. Equation (S6) is Eq. (S9) with $k_e = 0$, since the excess E_i is metal that the oxide does not accommodate at equilibrium, with the sweep velocity $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ as the interface velocity and v_i^{r} as the diffusive velocity of element i. The interface exchange diffusivities that would fix v_D are not known for the Fe–Cr–Ni oxide/metal interface (Yu et al. 2018), so the redistribution velocities are treated as element-specific parameters; the element that sets $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ (Cr under the present conditions) has $E_i = 0$ and is unaffected by its value.

S2.4 Comparison with the line scans

With the calculated oxide formed first placed at the centre of the intergranular oxide, consistent with symmetric growth into both grains, the revised model reproduces the central minimum of the Cr profile in the irradiated specimens and its deepening with dose, together with the measured dose dependence of the Cr and Fe contents of the oxide (Fig. 4 of the main text).

S3 Cross-Section-Averaged Oxygen Transport and the Lateral Closure

S3.1 Purpose and result

The grain-boundary oxidation model reduces oxygen transport to one dimension along the boundary. This note treats the entire oxide cross-section as the transport channel and derives, from the two-dimensional diffusion equation,

(i) the one-dimensional balance for the cross-section-averaged oxygen concentration $\bar{C}_{\mathrm{O}}$;

(ii) a closure that gives the interfacial flux J_{r} directly from $\bar{C}_{\mathrm{O}}$, the oxide half-width L, the diffusivity $D_{\mathrm{O}}^{\mathrm{eff}}$ and the interfacial reaction law $J_{\mathrm{r}}(C_{\mathrm{O}}^{\mathrm{int}})$.

The closure is obtained from the separable solution of the Laplace equation in the oxide cross-section. It reduces to one scalar equation for a dimensionless eigenvalue $\mu \in (0, \pi/2)$,

$$\boxed{\frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} = \sum_k J_{\mathrm{r}}^k\big(\bar{C}_{\mathrm{O}}\,\mu\cot\mu\big)} \tag{S10}$$

after which

$$\boxed{J_{\mathrm{r}} = \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L}, \qquad C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}\,\mu\cot\mu} \tag{S11}$$

Equation (S10) has a unique root and is solved pointwise by Newton's method, in the same way as the series-resistance closure of the earlier model was solved for $C_{\mathrm{O}}^{\mathrm{int}}$.

S3.2 Geometry and notation

- y: coordinate along the grain boundary, $y = 0$ at the surface.
- x: coordinate normal to the boundary, $x = 0$ on the boundary mid-plane (plane of symmetry). Only the half-domain $0 \le x \le L$ is considered.
- $L(y,t)$: distance from the mid-plane to the oxide/metal interface. After the original boundary channel is filled.
- $C_{\mathrm{O}}(x,y,t)$: concentration of mobile (dissolved) oxygen in the oxide.
- $D_{\mathrm{O}}^{\mathrm{eff}}(y,t)$: effective oxygen diffusivity for transport across the section, $D_{\mathrm{O}}^{\mathrm{eff},x}$ of the main text, the harmonic mean of the diffusivity profile across the oxide. The lateral problem is solved for an equivalent uniform oxide with the same series resistance; the balance along the boundary uses the arithmetic mean $D_{\mathrm{O}}^{\mathrm{eff},y}$.

- Cross-section average, mid-plane value and interface value:

$$\bar{C}_{\mathrm{O}}(y,t) = \frac{1}{L}\int_0^L C_{\mathrm{O}}\,\mathrm{d}x, \qquad C_{\mathrm{O}}^{\mathrm{c}} = C_{\mathrm{O}}(0,y,t), \qquad C_{\mathrm{O}}^{\mathrm{int}} = C_{\mathrm{O}}(L,y,t). \tag{S12}$$

- Interfacial oxygen flux (per unit boundary area, positive toward the metal):

$$J_{\mathrm{r}}(y,t) = -D_{\mathrm{O}}^{\mathrm{eff}}\left.\frac{\partial C_{\mathrm{O}}}{\partial x}\right|_{x=L}. \tag{S13}$$

- Interfacial reaction law: the total consumption $J_{\mathrm{r}}(C_{\mathrm{O}}^{\mathrm{int}}) = \sum_k J_{\mathrm{r}}^k(C_{\mathrm{O}}^{\mathrm{int}})$ of the rate laws of the main text. It is a nondecreasing function of $C_{\mathrm{O}}^{\mathrm{int}}$ at fixed metal concentrations. Where a linearized form is used below, $k \equiv \mathrm{d}J_{\mathrm{r}}/\mathrm{d}C_{\mathrm{O}}^{\mathrm{int}}$.

S3.3 Assumptions

A1. Two-dimensional Fickian diffusion of oxygen in an equivalent uniform oxide: the oxide of the main text, whose diffusivity varies across the section, is replaced, for the lateral problem, by a uniform oxide with the across-section diffusivity $D_{\mathrm{O}}^{\mathrm{eff}} = D_{\mathrm{O}}^{\mathrm{eff},x}$, the harmonic mean of the profile, which carries the same steady flux across the section.

A2. Symmetry at the mid-plane: $\partial C_{\mathrm{O}}/\partial x = 0$ at $x = 0$.

A3. Slow variation along the boundary of the half-width, the diffusivity and the metal concentrations that enter the reaction law: $|\partial L/\partial y| \ll 1$, and $D_{\mathrm{O}}^{\mathrm{eff}}$, $J_{\mathrm{r}}(\cdot)$ change little over a distance L.

A4. Lateral quasi-steady state: the lateral profile adjusts on the time scale $L^2/D_{\mathrm{O}}^{\mathrm{eff}}$, which is short compared with the time scales on which $\bar{C}_{\mathrm{O}}$ and L change.

A5. Locally, the lateral profile is the fundamental separable mode of the Laplace equation (Sec. S3.5). Higher modes decay over a distance of order $0.3L$ or less and are neglected (Sec. S3.7).

No assumption is made on whether the interface is reaction- or diffusion-limited, and no assumption is made that C_{O} is uniform across the section.

S3.4 Cross-section integration of the 2D balance

The two-dimensional conservation equation in the oxide is

$$\frac{\partial C_{\mathrm{O}}}{\partial t} = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y} \right) + D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial^2 C_{\mathrm{O}}}{\partial x^2}. \tag{S14}$$

Integrate each term over $0 \le x \le L(y,t)$, using the Leibniz rule where the limit depends on y or t.

Lateral term. With A2 and the definition (S13),

$$\int_0^L D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial^2 C_{\mathrm{O}}}{\partial x^2}\,\mathrm{d}x = D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial x}\bigg|_0^L = -J_{\mathrm{r}}. \tag{S15}$$

Time term. Because the upper limit moves, the Leibniz rule gives the identity

$$\frac{\partial}{\partial t}\int_0^L C_{\mathrm{O}}\,\mathrm{d}x = \int_0^L \frac{\partial C_{\mathrm{O}}}{\partial t}\,\mathrm{d}x + C_{\mathrm{O}}(L)\,\frac{\partial L}{\partial t}, \qquad \text{i.e.} \qquad \int_0^L \frac{\partial C_{\mathrm{O}}}{\partial t}\,\mathrm{d}x = \frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) - C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial t}. \tag{S16}$$

The last term is the dissolved oxygen, at concentration $C_{\mathrm{O}}^{\mathrm{int}}$, contained in the oxide formed during $\mathrm{d}t$.

Along-boundary term. Applying the Leibniz rule twice,

$$\int_0^L \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y} \right)\mathrm{d}x = \frac{\partial}{\partial y}\int_0^L D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y}\,\mathrm{d}x - D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y}\bigg|_{x=L} \frac{\partial L}{\partial y}, \tag{S17}$$

$$\int_0^L D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y}\,\mathrm{d}x = D_{\mathrm{O}}^{\mathrm{eff}} \left[\frac{\partial}{\partial y}\big(L\,\bar{C}_{\mathrm{O}}\big) - C_{\mathrm{O}}^{\mathrm{int}} \frac{\partial L}{\partial y} \right] = D_{\mathrm{O}}^{\mathrm{eff}}\, L\, \frac{\partial \bar{C}_{\mathrm{O}}}{\partial y} + D_{\mathrm{O}}^{\mathrm{eff}}\,(\bar{C}_{\mathrm{O}} - C_{\mathrm{O}}^{\mathrm{int}})\,\frac{\partial L}{\partial y}. \tag{S18}$$

Hence

$$\int_0^L \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y} \right)\mathrm{d}x = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}}\, L\, \frac{\partial \bar{C}_{\mathrm{O}}}{\partial y} \right) + \underbrace{\frac{\partial}{\partial y}\left[D_{\mathrm{O}}^{\mathrm{eff}} (\bar{C}_{\mathrm{O}} - C_{\mathrm{O}}^{\mathrm{int}}) \frac{\partial L}{\partial y} \right] - D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y}\bigg|_{x=L} \frac{\partial L}{\partial y}}_{O(\partial L/\partial y)}. \tag{S19}$$

Result. Collecting (S15)–(S19),

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}\right) - J_{\mathrm{r}} + C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial t} + O\!\left(\frac{\partial L}{\partial y}\right). \tag{S20}$$

Equation (S20) is exact. Under A3 the last group is dropped.

At the moving interface the diffusive flux J_{r} of (S13) supplies both the reaction and the dissolved oxygen in the fresh oxide (Stefan condition, dissolved oxygen in the metal neglected):

$$J_{\mathrm{r}} = \sum_k J_{\mathrm{r}}^k(C_{\mathrm{O}}^{\mathrm{int}}) + C_{\mathrm{O}}^{\mathrm{int}}\,\frac{\partial L}{\partial t}. \tag{S21}$$

Substituting (S21) into (S20), the terms $C_{\mathrm{O}}^{\mathrm{int}}\,\partial L/\partial t$ cancel and the sink is the reaction consumption alone. The working one-dimensional equation is

$$\boxed{\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}\right) - \sum_k J_{\mathrm{r}}^k} \tag{S22}$$

With $\partial L/\partial t = \Omega \sum_k J_{\mathrm{r}}^k$ and $\Omega = 1/C_{\mathrm{O}}^{\mathrm{oxide}}$ the oxide volume formed per unit oxygen consumed, the ratio of the Stefan term in (S21) to the reaction term is $C_{\mathrm{O}}^{\mathrm{int}}/C_{\mathrm{O}}^{\mathrm{oxide}} \ll 1$. The diffusive flux J_{r} and the reaction consumption $\sum_k J_{\mathrm{r}}^k$ therefore differ by a negligible amount, and from here on the single symbol J_{r} is used for both; in particular the closure of Sec. S3.6 equates J_{r} of (S13) to $\sum_k J_{\mathrm{r}}^k$. Two remarks:

- The along-boundary conductance is $D_{\mathrm{O}}^{\mathrm{eff}} L$ with the *same* $D_{\mathrm{O}}^{\mathrm{eff}}$ as in the lateral direction. This follows from A1 alone; no assumption on the lateral profile was used.
- L sits inside the time derivative. Expanding, $L\,\partial \bar{C}_{\mathrm{O}}/\partial t = \partial_y(D_{\mathrm{O}}^{\mathrm{eff}} L\,\partial_y \bar{C}_{\mathrm{O}}) - J_{\mathrm{r}} - \bar{C}_{\mathrm{O}}\,\partial L/\partial t$; the last term is the dilution of dissolved oxygen as the section widens.

Equation (S22) contains two unknowns per position, $\bar{C}_{\mathrm{O}}$ and J_{r}. The relation between them is derived next.

S3.5 Lateral problem: separation of variables

S3.5.1 Local problem

Under A3 and A4, in the neighbourhood of a given position along the boundary, the oxide occupies a half-strip $0 \le x \le L$ with L, $D_{\mathrm{O}}^{\mathrm{eff}}$ and the reaction law treated as constants, and the concentration satisfies the steady equation

$$\frac{\partial^2 C_{\mathrm{O}}}{\partial x^2} + \frac{\partial^2 C_{\mathrm{O}}}{\partial y^2} = 0, \qquad \left.\frac{\partial C_{\mathrm{O}}}{\partial x}\right|_{x=0} = 0, \qquad -D_{\mathrm{O}}^{\mathrm{eff}} \left.\frac{\partial C_{\mathrm{O}}}{\partial x}\right|_{x=L} = J_{\mathrm{r}}\big(C_{\mathrm{O}}(L,y)\big). \tag{S23}$$

S3.5.2 Separable solutions

Seek $C_{\mathrm{O}} = X(x)\,Y(y)$. Substituting into the Laplace equation and dividing by XY,

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\left(\frac{\mu}{L}\right)^2, \tag{S24}$$

where the separation constant is written as $-(\mu/L)^2$ with $\mu > 0$, because the x-problem must be oscillatory (it has a homogeneous Neumann condition at $x = 0$ and a sink at $x = L$) and the y-dependence must decay into the boundary. Then

$$X'' + \left(\frac{\mu}{L}\right)^2 X = 0, \quad X'(0) = 0 \;\Rightarrow\; X = \cos\!\left(\frac{\mu x}{L}\right); \qquad Y'' - \left(\frac{\mu}{L}\right)^2 Y = 0,\; Y \to 0 \;\Rightarrow\; Y = e^{-\mu y/L}. \tag{S25}$$

Hence

$$C_{\mathrm{O}}(x,y) = A\,\cos\!\left(\frac{\mu x}{L}\right) e^{-\mu y/L}, \tag{S26}$$

where A is an amplitude and μ is to be fixed by the interfacial condition. Different values of μ give different modes; the fundamental mode has the smallest μ and the slowest decay in y, and by A5 it is the one retained.

S3.5.3 Averages and flux of the mode

From (S26), with the common factor $e^{-\mu y/L}$ absorbed into A,

$$\bar{C}_{\mathrm{O}} = \frac{A}{L}\int_0^L \cos\!\Big(\frac{\mu x}{L}\Big)\,\mathrm{d}x = A\,\frac{\sin\mu}{\mu}, \tag{S27}$$

$$C_{\mathrm{O}}^{\mathrm{int}} = A\cos\mu, \tag{S28}$$

$$C_{\mathrm{O}}^{\mathrm{c}} = A, \tag{S29}$$

$$J_{\mathrm{r}} = -D_{\mathrm{O}}^{\mathrm{eff}}\,\frac{\partial C_{\mathrm{O}}}{\partial x}\bigg|_{x=L} = \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,A\,\mu\sin\mu}{L}. \tag{S30}$$

Eliminating the amplitude with (S27), $A = \mu\bar{C}_{\mathrm{O}}/\sin\mu$, gives the two relations of (S11):

$$J_{\mathrm{r}} = \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L}, \qquad C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}\,\mu\cot\mu, \qquad C_{\mathrm{O}}^{\mathrm{c}} = \frac{\mu\,\bar{C}_{\mathrm{O}}}{\sin\mu}. \tag{S31}$$

Both J_{r} and $C_{\mathrm{O}}^{\mathrm{int}}$ are expressed through $\bar{C}_{\mathrm{O}}$ and μ only. Note that $\mu\cot\mu$ decreases monotonically from 1 at $\mu = 0$ to 0 at $\mu = \pi/2$, so $0 \le C_{\mathrm{O}}^{\mathrm{int}} \le \bar{C}_{\mathrm{O}}$ for $\mu \in [0, \pi/2]$.

S3.5.4 Consistency with the averaged equation

Substituting the mode into the steady form of (S22) with constant L and $D_{\mathrm{O}}^{\mathrm{eff}}$,

$$D_{\mathrm{O}}^{\mathrm{eff}}\,L\,\frac{\partial^2\bar{C}_{\mathrm{O}}}{\partial y^2} = D_{\mathrm{O}}^{\mathrm{eff}}\,L\left(\frac{\mu}{L}\right)^2\bar{C}_{\mathrm{O}} = \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} = J_{\mathrm{r}}, \tag{S32}$$

with no residual. The averaged equation and the mode closure are therefore mutually consistent; no additional approximation is introduced by combining them.

S3.6 The equation for μ

S3.6.1 General (nonlinear) reaction law

Imposing the interfacial condition of (S23) on the mode, and using (S31) to express both sides through $\bar{C}_{\mathrm{O}}$ and μ,

$$g(\mu) \equiv \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} - J_{\mathrm{r}}\big(\bar{C}_{\mathrm{O}}\,\mu\cot\mu\big) = 0. \tag{S33}$$

This is Eq. (S10). It is a single scalar equation in μ; the inputs are the local values of $\bar{C}_{\mathrm{O}}$, L, $D_{\mathrm{O}}^{\mathrm{eff}}$ and the metal concentrations that enter $J_{\mathrm{r}}(\cdot)$.

Existence and uniqueness. On $(0, \pi/2)$ the first term increases monotonically from 0; the second term decreases monotonically, because $J_{\mathrm{r}}(\cdot)$ is nondecreasing and $\mu \cot \mu$ is decreasing. Hence g is strictly increasing, with

$$g(0^+) = -J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}), \qquad g(\pi/2) = \frac{\pi^2 D_{\mathrm{O}}^{\mathrm{eff}}\, \bar{C}_{\mathrm{O}}}{4L} - J_{\mathrm{r}}(0). \tag{S34}$$

If $J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}) > 0$ (oxidation possible at the local average concentration) and $J_{\mathrm{r}}(0) \le 0$ (no consumption at zero oxygen, which holds for all three rate laws), there is exactly one root $\mu^* \in (0, \pi/2)$. If $J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}) \le 0$, no oxygen is consumed: set $\mu = 0$, $J_{\mathrm{r}} = 0$, $C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}$.

Newton iteration. The derivative is

$$g'(\mu) = \frac{2 D_{\mathrm{O}}^{\mathrm{eff}}\, \mu\, \bar{C}_{\mathrm{O}}}{L} - J_{\mathrm{r}}'\big(C_{\mathrm{O}}^{\mathrm{int}}\big)\, \bar{C}_{\mathrm{O}}\, \frac{\mathrm{d}}{\mathrm{d}\mu}\big(\mu \cot \mu\big), \qquad \frac{\mathrm{d}}{\mathrm{d}\mu}\big(\mu \cot \mu\big) = \frac{\sin\mu \cos\mu - \mu}{\sin^2 \mu} < 0, \tag{S35}$$

so $g' > 0$ throughout and Newton's method converges from any bracketed start. A safeguarded Newton (fall back to bisection if an iterate leaves $(0, \pi/2)$) is used. A good initial guess is the linearized value of Sec. S3.6.2.

Algorithm per grid point and time step.

1. Inputs: $\bar{C}_{\mathrm{O}}$, L, $D_{\mathrm{O}}^{\mathrm{eff}}$, $C_{\mathrm{Cr}}^{\mathrm{GB}}$, $C_{\mathrm{Fe}}^{\mathrm{GB}}$, $C_{\mathrm{Si}}^{\mathrm{GB}}$.
2. If $J_{\mathrm{r}}(\bar{C}_{\mathrm{O}}) \le 0$: $\mu = 0$, $J_{\mathrm{r}} = 0$, $C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}$. Stop.
3. Solve $g(\mu) = 0$ on $(0, \pi/2)$ by safeguarded Newton.
4. $J_{\mathrm{r}} = D_{\mathrm{O}}^{\mathrm{eff}} \mu^2 \bar{C}_{\mathrm{O}} / L$; $C_{\mathrm{O}}^{\mathrm{int}} = \bar{C}_{\mathrm{O}}\, \mu \cot \mu$; the individual $J_{\mathrm{r}}^k(C_{\mathrm{O}}^{\mathrm{int}})$ give the metal sinks, the stoichiometric sinks of the main text, and the growth of each L_k.
5. Use J_{r} as the sink in (S22).

This replaces the series-resistance closure of the earlier model, which also solves one scalar equation per point; the unknown is now μ instead of $C_{\mathrm{O}}^{\mathrm{int}}$.

S3.6.2 Linearized reaction law

For $J_\mathrm{r} = k\, C_\mathrm{O}^\mathrm{int}$, Eq. (S33) becomes

$$\frac{D_\mathrm{O}^\mathrm{eff}\mu^2}{L} = k\,\mu\cot\mu \qquad \Longleftrightarrow \qquad \mu\tan\mu = \frac{kL}{D_\mathrm{O}^\mathrm{eff}} \equiv \mathrm{Da}, \tag{S36}$$

the classical eigenvalue condition for a Robin boundary. Da is the ratio of the lateral diffusion resistance $L/D_\mathrm{O}^\mathrm{eff}$ to the reaction resistance $1/k$. The concentration $\bar{C}_\mathrm{O}$ drops out: μ depends only on Da.

$\mathrm{Da} = kL/D_\mathrm{O}^\mathrm{eff}$	μ	$C_\mathrm{O}^\mathrm{int}/\bar{C}_\mathrm{O} = \mu\cot\mu$	$J_\mathrm{r}L/(D_\mathrm{O}^\mathrm{eff}\bar{C}_\mathrm{O}) = \mu^2$
0	0	1	0
0.1	0.311	0.968	0.097
1	0.860	0.740	0.740
10	1.429	0.204	2.042
∞	$\pi/2$	0	$\pi^2/4 \approx 2.467$

Table S1: Fundamental mode for a linear reaction law.

S3.6.3 Limits

- **Reaction-limited**, $\mathrm{Da} \ll 1$: $\mu^2 \simeq \mathrm{Da}$, so $J_\mathrm{r} \simeq k\,\bar{C}_\mathrm{O}$ and $C_\mathrm{O}^\mathrm{int} \simeq \bar{C}_\mathrm{O}$. The interface sees the average concentration; the lateral profile is nearly uniform, with a parabolic deviation of relative size $O(\mathrm{Da})$. The decay length along the boundary is $L/\mu \simeq \sqrt{D_\mathrm{O}^\mathrm{eff}L/k} \gg L$.
- **Diffusion-limited**, $\mathrm{Da} \gg 1$: $\mu \to \pi/2$, so $J_\mathrm{r} \to \pi^2 D_\mathrm{O}^\mathrm{eff}\bar{C}_\mathrm{O}/(4L)$ and $C_\mathrm{O}^\mathrm{int} \to 0$. Oxygen is consumed as soon as it reaches the interface; the lateral profile is a quarter cosine; the decay length along the boundary is $2L/\pi \approx 0.64L$.

Between these limits μ interpolates smoothly. The closure never needs a shape factor: if one insists on writing $J_\mathrm{r} = f\, D_\mathrm{O}^\mathrm{eff}(\bar{C}_\mathrm{O} - C_\mathrm{O}^\mathrm{int})/L$, then $f = \mu^2/(1 - \mu\cot\mu)$, which runs from 3 to $\pi^2/4$, but f plays no role in the algorithm.

S3.7 Range of validity

1. **Higher modes.** The interfacial condition admits a sequence of eigenvalues $\mu_0 < \mu_1 < \ldots$, with $\mu_n \in (n\pi,\, n\pi + \pi/2)$ for a Robin condition. Each decays as $e^{-\mu_n y/L}$. The first higher mode decays at least as fast as $e^{-\pi y/L}$, i.e. over a distance $\lesssim L/\pi \approx 0.3L$, compared with $\geq 2L/\pi$ for the fundamental. Higher modes are excited where the boundary data change abruptly: the surface entry, the oxidation tip, and any position where L, $D_{\mathrm{O}}^{\mathrm{eff}}$ or the metal concentrations jump. Away from such places, only the fundamental survives (A5).

2. **Slowly varying geometry (A3).** The mode is derived for a strip of constant width. For a wedge-shaped oxide the closure is applied with the local $L(y)$; the error is $O(\partial L/\partial y)$, the same order as the terms dropped in (S20).

3. **Quasi-steady lateral profile (A4).** Requires $L^2/D_{\mathrm{O}}^{\mathrm{eff}}$ to be short compared with the time scales of $\bar{C}_{\mathrm{O}}(t)$ and $L(t)$. This can be checked from the simulation output.

4. **Early stage.** When the oxidized zone is as deep as it is wide, the decay length is comparable to L and the whole problem is two-dimensional. The one-dimensional description is meaningful only once the oxidized zone is slender; the initial aspect-ratio-one stage lies outside its scope.

5. **Equivalent uniform oxide (A1).** The diffusivity of the oxide varies across the section, and the lateral problem is solved with the harmonic mean $D_{\mathrm{O}}^{\mathrm{eff},x}$, which is exact for a one-dimensional steady flux across the section. For a nonuniform $D_{\mathrm{O}}^{\mathrm{eff}}(x)$ the fundamental mode is not a pure cosine; the deviation is neglected. The conductance along the boundary in (S22) uses the arithmetic mean $D_{\mathrm{O}}^{\mathrm{eff},y}$, exact for flux along the direction in which $D_{\mathrm{O}}^{\mathrm{eff}}$ is uniform; the two means are the Wiener bounds of effective-medium theory.

S3.8 Summary of the model change

Earlier formulation:

$$\frac{\partial C_{\mathrm{O}}}{\partial t} = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}} \frac{\partial C_{\mathrm{O}}}{\partial y} \right) - \frac{J_{\mathrm{r}}}{\delta_{\mathrm{GB}}}, \quad J_{\mathrm{r}} = D_{\mathrm{O}}^{\mathrm{eff}} \frac{C_{\mathrm{O}} - C_{\mathrm{O}}^{\mathrm{int}}}{\sum_k L_k} = \sum_k J_{\mathrm{r}}^k(C_{\mathrm{O}}^{\mathrm{int}}). \quad \text{(S37)}$$

Proposed formulation (cross-section average over the whole oxide, fundamental-mode closure):

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = \frac{\partial}{\partial y}\left(D_{\mathrm{O}}^{\mathrm{eff}}\,L\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}\right) - J_{\mathrm{r}}, \quad J_{\mathrm{r}} = \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L}, \quad \frac{D_{\mathrm{O}}^{\mathrm{eff}}\,\mu^2\,\bar{C}_{\mathrm{O}}}{L} = \sum_k J_{\mathrm{r}}^k\big(\bar{C}_{\mathrm{O}}\,\mu\cot\mu\big). \tag{S38}$$

What changes: (i) the storage width and along-boundary conductance are the time-dependent half-width L instead of the fixed δ_{GB}, with L inside the time derivative; (ii) the lateral closure is the fundamental Laplace mode, valid from the reaction-limited to the diffusion-limited regime, with μ obtained pointwise by Newton's method; (iii) L is the mid-plane-to-interface distance. What does not change: the reaction laws, the metal sinks, the definition of $D_{\mathrm{O}}^{\mathrm{eff}}$ as a local volume-fraction average, and the structure of the pointwise scalar solve.

S4 Apparent Effective Diffusivities of the Oxidation Front

Table S2 lists the apparent effective diffusivities $\bar{D}_{\mathrm{eff}} = L^2/2t$ computed from the reported final penetration depth L and total exposure time t of the two experimental series used in the main text. The values assume parabolic kinetics from $t = 0$ with no incubation correction. At the same dose the oxide penetrates faster in the series of Deng et al. (2020) than in that of Deng et al. (2017); the 0.5 dpa value of the former exceeds the 3.0 dpa value of the latter. A model calibrated on the 2017 series is therefore expected to underestimate the absolute depths of the 2020 series, as observed in Fig. 8 of the main text. The normalized depth used there is

$$L_{\mathrm{norm}} = \frac{L}{L_{\mathrm{max}}}, \tag{S39}$$

where L_{max} is the maximum penetration depth over both doses, taken separately for the calculations (110 nm) and for the measurements (231 nm).

Table S2: Apparent effective diffusivities of the grain-boundary oxidation front, $\bar{D}_{\text{eff}} = L^2/2t$, from the reported final penetration depth L and total exposure time t. Depths for Deng et al. (2017) are digitized from Fig. 12 therein (mean over the reported boundaries); depths for Deng et al. (2020) are the values reported in the text.

Source	Dose	t (h)	L (nm)	$\bar{D}_{\text{eff}}$ (nm^2/h)	$\bar{D}_{\text{eff}}$ (10^{-22} m^2/s)
Deng et al. (2017)	SA	500	40	1.6	4.4
Deng et al. (2017)	0.5 dpa	500	60	3.6	10.0
Deng et al. (2017)	3.0 dpa	500	100	10.0	27.8
Deng et al. (2020)	0.5 dpa	1500	190	12.0	33.4
Deng et al. (2020)	1.5 dpa	1500	240	19.2	53.3

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